

Objective

“Evaluate whether AI tools are improving student academic performance, productivity, and confidence—or creating risks like dependency and ethical concerns.”

Key Insight:

1. AI Usage Patterns

- How frequently students use AI (Task_Frequency_Daily)
- If high concern students are still using it, means they are neglecting their privacy for sake of work and bad use of AI.
- Most common tools (Primary_AI_Tool)
- Main usage purposes (Main_Usage_Case)
- Differences across majors and age groups
- Productivity increased or decreased, means people who use AI more are more confident or less confident about their career

Goal: Identify who uses AI and how

2. Academic Impact (Core KPI)

- Compare GPA_Baseline vs GPA_Post_AI
- Measure improvement or decline
- How often do they use AI tools and their concern regarding this?
- If GPA increases → positive academic impact
- If decreases → potential over-reliance or misuse

3. Productivity & Time Efficiency

- Weekly time saved (Time_Saved_Hours_Weekly)
- Compare with task frequency
- Key Question:
- Are frequent users actually saving more time?

4. Career Confidence Impact

Career_Confidence_Score Relation with:

- GPA improvement
- AI usage frequency

Insight:

- Does AI build confidence or reduce independent thinking?
- Is student career confidence related to their increased cgpa?

5. Ethical Concerns (Critical Risk Indicator)

- AI_Ethics_Concern Interpretation:
- Low → comfortable using AI
- High → worried about privacy, bias, etc.

Important patterns:

- High concern + high usage = risky behavior
- Tool-wise concern comparison

6. Dependency vs Benefit

- High usage + low GPA improvement = dependency risk
- High usage + high improvement = effective use
- High CGPA+ High career confidence= Risky

7. Student Segmentation

Benefiting Students

- GPA increased
- Time saved
- Confidence increased
- Low ethical concern

At-Risk Students

- No GPA improvement
- High usage
- High ethical concern

- Low confidence

8. Key Relationships to Analyze

- Age vs Ethical Concern
- Age vs Confidence
- Usage vs Time Saved
- Usage vs GPA Change
- GPA Change vs Confidence
- Tool vs Ethical Concern
- Usage Case vs Time Saved

AI Impact on Student Life – Final Report

1. How students are using AI

Students are using AI mostly for:

- Exam preparation
- Brainstorming ideas
- Writing and literature review
- Understanding concepts

Most used tools are GPT and Claude, followed by Gemini, Perplexity, and Copilot.

Simple insight

Students are not just using AI for small help. They are using it for important thinking tasks like writing and studying.

This means AI is becoming part of their learning process, not just a helper tool.

2. Ethical concerns

- Low concern: 34 percent

- Medium concern: 32 percent
- High concern: 33 percent

Age group 20 shows the highest concern.

Simple insight

Students know AI has risks, but they still use it anyway.

So awareness is there, but it is not changing behavior.

3. Academic performance (GPA)

- Correlation between AI usage and GPA is almost zero (0.009)
- Some students improve, some decline, many stay the same

Simple insight

AI is NOT clearly improving or damaging grades overall.

It is mixed and not consistent.

4. Productivity and time saving

- Students using AI less and more often get almost the same results
- Time saved is also almost the same for both groups

Simple insight

Using AI more does NOT mean you save more time or perform better.

It depends on how you use it, not how much you use it.

5. Career confidence

- Average confidence score is moderate (around 5 out of 10)
- No strong link between GPA improvement and confidence increase

Simple insight

Even if students perform better, they do not necessarily feel more confident.

Many students may still feel unsure about their future.

6. Who is actually benefiting

- Benefiting students: 4.33 percent
- Non-benefiting students: 95.67 percent

Simple insight

Only a very small number of students are actually getting real benefit from AI.

Most students are using AI, but not gaining strong measurable improvement.

7. Important behavioral finding

- Students using AI 1 time or 7 times per day show similar results
- Time saved is also almost the same

Simple insight

More usage does NOT mean better results.

What matters is how they use AI, not how often.

Final conclusion

AI is widely used by students, especially for studying and exam preparation. However, its real impact is not strong or consistent.

Here is the simple truth from the data:

- AI does not clearly improve GPA
- Only a small group of students benefit
- Most students show no major change
- Ethical concerns exist but do not stop usage
- Using AI more does not guarantee better results or more time saving

AI is helpful, but not equally effective for all students. Its impact depends more on how students use it rather than how often they use it.

AI is widely adopted across students for academic and cognitive tasks including exam preparation, brainstorming, and research activities. However, the data shows that its impact is not uniformly beneficial.

There is no significant improvement in academic performance associated with AI usage. Only a very small proportion of students experience meaningful benefits. Ethical concerns are present across all usage levels, and increased usage does not necessarily lead to better outcomes or higher confidence.

Overall, AI in education currently represents a high adoption but low impact system, where benefits are limited to a small subset of users and outcomes depend heavily on usage behavior rather than tool availability.